

SAM MAHDAVIAN, PhD, PMP, PMI-PBA

AI Solutions & Delivery Lead · Founder & CEO, Brane

Agentic systems that solve the business problem. Tested, governed, and safe in regulated production.

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PROFESSIONAL SUMMARY

AI solutions leader who turns ambiguous business problems into agentic systems that hold up in regulated production. I work as a single-threaded owner from discovery and pre-sales through reference architecture, build, and adoption, and every system I ship has to clear one test: it solves the business problem, it is evaluated against real cases, and it is safe to run. Across healthcare, financial services, transportation, retail, and manufacturing I have shipped 40+ systems and 150+ agents that together produced \$40M+ in revenue and \$500M+ in identified client savings, and I have authored 50+ Claude skills that specialize those agents into repeatable, gated engines. I also co-founded and run Brane, an AI siting-and-planning startup, as CEO. For 15+ years I have been the translator between the C-suite that wants ROI and the engineers who build, pairing LLM/RAG and AWS engineering with executive-facing delivery. PhD (ML / applied AI); PMP; PMI-PBA.

CORE COMPETENCIES

Solutions & Pre-Sales: enterprise AI delivery, solution architecture, discovery to design to deploy to adopt, requirements elicitation across sectors, pre-sales and POCs, buyer-persona and commercial market analysis, ROI and business-case framing, single-threaded ownership, executive enablement.

Agentic AI & GenAI: LangChain/LangGraph, Strands SDK, CrewAI, Claude Code and custom Python; multi-agent orchestration; agent specialization through 50+ authored skills (workflow, capability, domain) with approval gates and append-only decision logs as agent memory; MCP tool design; RAG and vector DBs; guardrails, evals, and human-in-the-loop; prompt engineering; latency/cost/accuracy optimization; AWS Bedrock and AgentCore; Claude and OpenAI.

Production & MLOps: AWS (Bedrock, SageMaker, Lambda, Step Functions, S3, ECS/EKS), Docker, Kubernetes, CI/CD, IaC (Terraform/CDK), model serving and LLMops, observability, model-risk governance, IAM and cloud security, HITRUST/HIPAA, FinOps.

Leadership & Delivery: team building and mentorship (orgs to ~25), roadmap and solution ownership, GTM, pricing and packaging, Agile/Scrum and SAFe, storytelling with data, hiring and talent development.

PROFESSIONAL EXPERIENCE

AArete LLC Director, Data Science & AI

Orlando, FL · 2022 - Present

- Designed the architecture and led the build of Doczy.ai, a multi-tenant GenAI platform on AWS (retrieval, prompt orchestration, agentic tool-calling, dynamic model selection, governance, observability). Scaled it to 2.5M+ contracts and 442B+ tokens at 99%+ accuracy under HITRUST/HIPAA, generating \$15M+ in revenue and \$330M+ in client savings. Architecture featured on the AWS Architecture Blog; issued a U.S. patent in 2025.
- Built 40+ systems and 150+ agents that each solve a specific business problem, most now in production, on LangChain/LangGraph, Strands SDK, Claude Code, and custom Python over AWS Bedrock and AgentCore. Across the portfolio these systems produced \$40M+ in revenue and \$500M+ in identified client savings. Every one is evaluated against real cases before release, tuned for latency, cost, and accuracy, and built human-in-the-loop and audit-first.
- Authored 50+ Claude skills across 12 engine repositories that turn one-off agent work into repeatable, gated procedures: workflow skills (investigate, requirements with an approval gate, build, test), capability skills (PHI/PII pre-flight, engine repeatability, cost observability), and domain skills per engine. Result: any teammate regenerates a deliverable from a clean clone with one command, and a strict split between LLM judgment and deterministic precision is enforced by the tooling, not by discipline.
- Built an AI claims-testing automation engine for a Medicaid managed-care plan that derives what each claim should pay from contract text and the state's published rate files, with no LLM anywhere in the pricing chain: 394 traceable test cases from 102 contract rules, backtested against 213,420 historical claim lines, 460+ automated tests, regenerated in minutes from one command, roughly 100x faster than hand-written test authoring. Progressed the work from business development to a signed SOW.
- Run discovery and pre-sales as a single-threaded owner across 10+ Fortune 500 engagements. I sit with executives and business teams, map the real pain by sector, and convert ambiguous problems into structured SOWs and roadmaps. Recently won a \$700K+ engagement with an equipment-finance lender, plus \$2.5M+ in competitive government and industry RFPs.
- Own production deployment (Docker, ECS/EKS, CI/CD, IaC, Bedrock/SageMaker serving) and authored the firm's AI-governance framework: model-risk management, bias and drift monitoring, explainability, prompt-injection defense, confidence-scored human-in-the-loop, and a four-stage maturity model that gates every system before it ships.
- Lead and mentor ~15 AI scientists and engineers across backend, frontend, and DevOps, plus a business-translation layer that turns stakeholder needs into specs and prompts. Run the firm-wide AI delivery program: ~\$1.3M budget, 10+ concurrent workstreams, a six-gate value framework, and value-based pricing that protects margin as delivery cost falls.
- Lead executive AI-enablement workshops; one for 40+ leaders at an S&P 500 global manufacturer produced a 30% productivity increase and 20+ prioritized use cases. Earlier (Manager / Sr. Data Scientist), built an AI/ML Accelerator that cut modeling cycles ~75% and an ML suite proving a 4.1% registration uplift worth \$9.6M in new annual revenue.

Brane (SPLUSM LLC) Co-Founder & CEO

Orlando, FL · 2020 - Present

- Co-founded and lead Brane Mobility, a cloud-native B2B SaaS that turns raw mobility and urban data into multi-scenario, stakeholder-ready insights in minutes for city planners, public agencies, and enterprises. Built and led a ~25-person cross-functional org (GenAI and data scientists, data engineers, full-stack developers, UX/UI) and awarded \$625K in grants.
- In 2026 launched Brane AI on the same platform: a data-center siting and approval-intelligence product that screens every census block group in a three-county corridor against 40 indicators (six gating), prices nine mitigation levers with benefit-cost and lifecycle math, and serves developers and county, utility, and water-district reviewers from one scoring object. Every on-screen number reproduces through the product's own API, enforced by a six-layer test gate; nine roadmap phases shipped.
- Architected the platform as two integrated systems: a data-management layer (a four-level dataset-to-KPI hierarchy with automated ETL, validation, a custom KPI formula builder, and user-data integration) and a five-step analysis workflow (project definition across 18 problem types, hotspot identification, multi-objective optimization, sensitivity analysis, and automated Word, PDF, and slide reporting). The AI Decision Engine pairs neural-network forecasting and genetic-algorithm optimization with an AWS Bedrock copilot; multi-city pilots showed ~18% operational efficiency gains.

UCF College of Engineering & Computer Science Lead Data Scientist & Researcher

Orlando, FL · 2017 - 2020

- Delivered two production applications with end-to-end data pipelines used by the U.S. Department of Transportation for traffic and cost forecasting; secured competitive USDOT and NSF funding. Led applied-AI research in forecasting, optimization, and simulation, producing 20+ peer-reviewed publications.

Stratus International Contracting Co. Senior Data Analyst

· 2011 - 2017

- Built a PMO optimization framework (hybrid genetic algorithm plus constraint-based simulation) and stood up the firm's first PMI-aligned PMO, cutting 21% of cost and 32% of time across major projects in three years.

SELECTED PRODUCTION AI SOLUTIONS (BUSINESS OUTCOME FIRST)

- Cut payer-policy and claims analysis time 90% with a 10% accuracy gain for a top-5 health payer, surfacing millions in recoverable overpayments. A 9-agent LangGraph pipeline generates 200+ Snowflake-ready SQL queries per policy; reused across payers.
- Cut invoice processing time 92% with a 5% accuracy gain across an airline, a manufacturer, and a large health system with a straight-through invoice agent on AWS Bedrock (Textract plus validation), routing only exceptions to a human.
- Surfaced spend savings across 14,765 suppliers and 47,489 parts for a Fortune 500 airline: a 10-agent hand-orchestrated pipeline plus a React command center delivered a 98% task-time reduction and a 20% accuracy gain.
- SpendSphere: a browser-native spend-intelligence platform (React + DuckDB-WASM, zero external requests) over AP spend, 240K MRO parts lines, and payer vendor cost, with an agentic analyst that writes and runs real SQL and cites its figures; every insight is SQL-computed, never LLM-generated, and release is gated by 160+ browser checks driven by six skills.
- Skills-driven deck builder: eight authored skills take source files to a client-ready PowerPoint on the corporate template; deterministic ground truth traces every dollar, and a QC gate stops the build if an unhedged figure misses its metric by more than 2%. 80% less deck time, zero untraced numbers.
- Cut unit-cost claims analysis time 35% with a 4% accuracy gain for a state Medicaid program using a 7-agent LangGraph cognitive loop, taking a five-day analysis down to hours.
- Protected margin on thousands of daily financing deals for an equipment-finance lender. An 8-agent pricing engine (XGBoost risk scoring, reinforcement-learning optimization, SHAP explainability) is targeting a 15 to 20% margin improvement and 60% faster decisions, explainable for compliance, for a newly won \$700K+ engagement.
- Made underwriting transparent and regulator-ready with an 11-agent LangGraph system that scores 11 risk dimensions under a Chief Underwriter orchestrator (XGBoost + SHAP, every decision explained). Open source.
- Built contract review that ships itself: a 7-agent clause-assessment engine rates every clause P/A/D/U from a tool-called rubric (criteria never hard-coded), with deterministic guards, an independent QC agent, and human routing for anything uncertain; 85% of assessments validated to auto-approve.
- Caught the fraud rings rules miss with a 3-agent LangGraph investigation (graph neural net plus LLM) over a 3D transaction graph. Open source.

EDUCATION & CERTIFICATIONS

Ph.D., Engineering, University of Central Florida. Dissertation: machine-learning algorithms for forecasting (ML / applied-AI methods).

M.S., Engineering, UCF. Executive Education, Business Administration, Harvard Business School.

Certifications: PMP · PMI-PBA · Certified SAFe Practitioner · Advanced Data Science Fellowship (The Data Incubator) · Claude Certified Architect, Anthropic (expected 2026) · AWS Solutions Architect Associate (expected 2026).

PUBLICATIONS, PATENTS & RECOGNITION

Patent & features: U.S. Patent (2025), GenAI contract-intelligence method (Doczy.ai) · featured on the AWS Architecture Blog · founder of Modern AI Academy, teaching how agentic AI is built and shipped.

Research: 20+ peer-reviewed publications, 460 citations, h-index 10, i10-index 10 (Google Scholar, Sept 2026), in IEEE Access, MDPI, and the ASCE Journal of Construction Engineering & Management. Most-cited: drivers and barriers to connected, automated, shared, and electric vehicles (IEEE Access, 139 citations). Reviewer for the Journal of Intelligent Transportation Systems and TRB.